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JEE MAIN · QUESTION PAPER

JEE Main 2022 (Online) 27th June Morning Shift

JEE Main Question Paper with Answer Key

27 Jun 2022	90	23	English
HELD ON	QUESTIONS	OUT OF SYLLABUS	LANGUAGE

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Chemistry	19 questions
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Out of syllabus	23 questions
Answer key	at the end

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Chemistry 19 questions

Q1. MCQ (Single Correct Answer)

Open solution

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : At 10°C, the density of a 5 M solution of KCl [atomic masses of K & Cl are 39 & 35.5 g mol^{-1} respectively], is 'x' g ml^{-1}. The solution is cooled to $-21°C$. The molality of the solution will remain unchanged.

Reason (R) : The molality of a solution does not change with temperature as mass remains unaffected with temperature.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

Q2. MCQ (Single Correct Answer)

Open solution

Based upon VSEPR theory, match the shape (geometry) of the molecules in List-I with the molecules in List-II and select the most appropriate option.

	List - I (Shape)		List - II (Molecules)
(A)	T-shaped	(I)	XeF_4
(B)	Trigonal planar	(II)	SF_4
(C)	Square planar	(III)	ClF_3
(D)	See-saw	(IV)	BF_3

(A) (A) - (I), (B) - (II), (C) - (III), (D) - (IV)

(B) (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

(C) (A) - (III), (B) - (IV), (C) - (II), (D) - (I)

(D) (A) - (IV), (B) - (III), (C) - (I), (D) - (II)

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Q3. MCQ (Single Correct Answer)

Open solution

Match List-I with List-II.

	List - I		List - II
(A)	Spontaneous process	(I)	$\Delta H < 0$
(B)	Process with $\Delta P = 0$, $\Delta T = 0$	(II)	$\Delta G_{T,P} < 0$
(C)	$\Delta H_{reaction}$	(III)	Isothermal and isobaric process
(D)	Exothermic Process	(IV)	[Bond energies of molecules in reactants] − [Bond energies of product molecules

Choose the correct answer from the options given below :

(A) (A) - (III), (B) - (II), (C) - (IV), (D) - (I)

(B) (A) - (II), (B) - (III), (C) - (IV), (D) - (I)

(C) (A) - (II), (B) - (III), (C) - (I), (D) - (IV)

(D) (A) - (II), (B) - (I), (C) - (III), (D) - (IV)

Q4. MCQ (Single Correct Answer)

Open solution

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : The ionic radii of O^{2-} and Mg^{2+} are same.

Reason (R) : Both O^{2-} and Mg^{2+} are isoelectronic species.

In the light of the above statements, choose the correct answer from the options given below.

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

Q5. MCQ (Single Correct Answer)

Open solution

Which of the following will have maximum stabilization due to crystal field?

(A) $[Ti(H_2O)_6]^{3+}$ **(B)** $[Co(H_2O)_6]^{2+}$ **(C)** $[Co(CN)_6]^{3-}$ **(D)** $[Cu(NH_3)_4]^{2+}$

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Q6. MCQ (Single Correct Answer)

Open solution

'A' and 'B' respectively are:

$$A \xrightarrow[(2)\ Zn - H_2O]{(1)\ O_3} \text{Ethane-1,2-dicarbaldehyde + Glyoxal/Oxaldehyde}$$

$$B \xrightarrow[(2)\ Zn - H_2O]{(1)\ O_3} \text{5-oxohexanal}$$

(A) 1-methylcyclohex-1,3-diene & cyclopentene.

(B) Cyclohex-1,3-diene & cyclopentene

(C) 1-methylcyclohex-1,4-diene & 1-methylcyclopent-1-ene

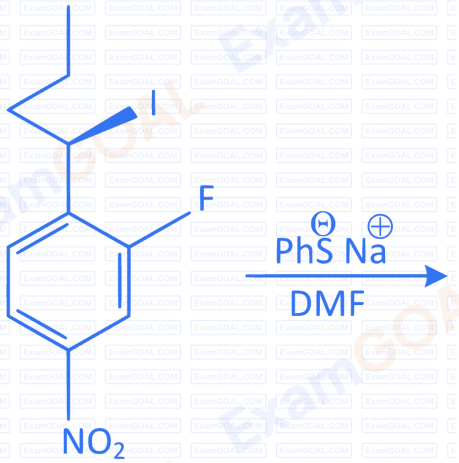
(D) Cyclohex-1,3-diene & 1-methylcyclopent-1-ene

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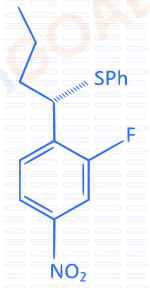
Q7. MCQ (Single Correct Answer)

Open solution

The major product of the following reaction is :



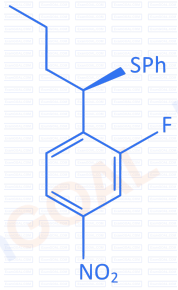
(A)



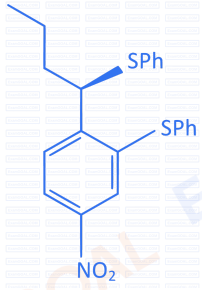
(B)



(C)



(D)

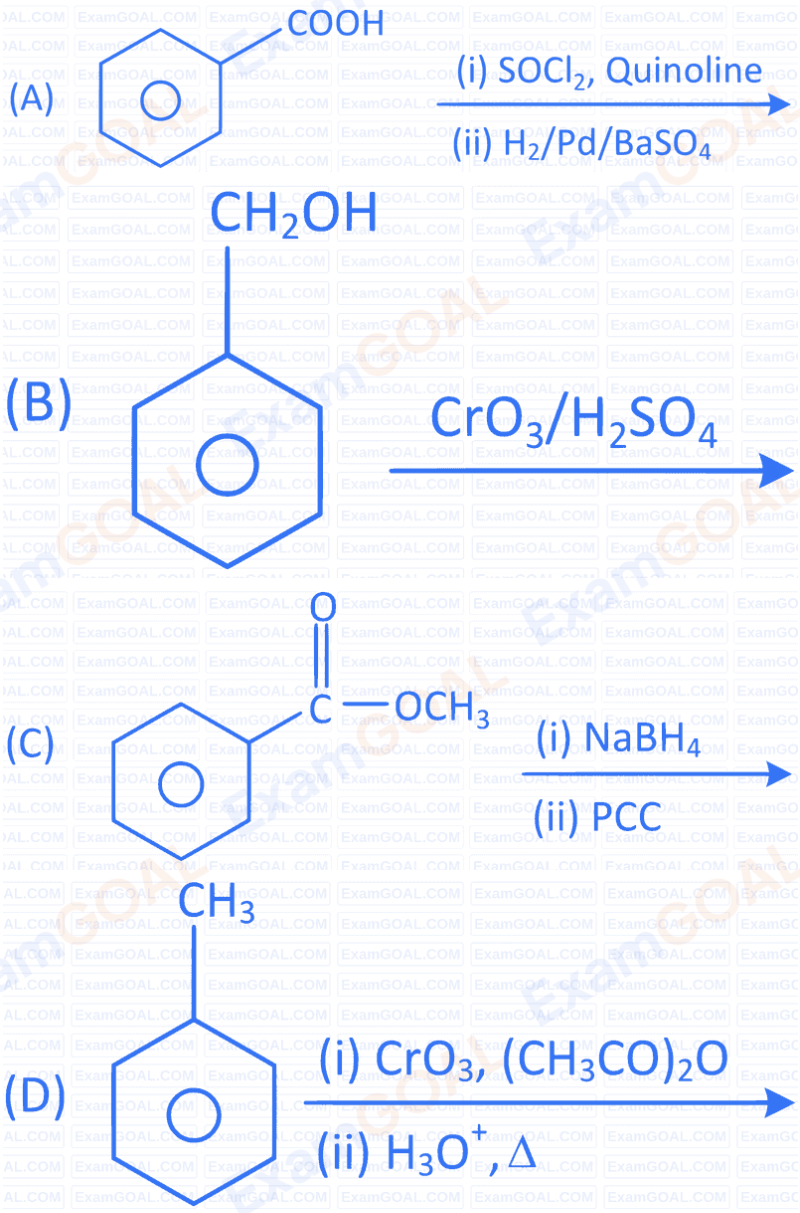


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Q8. MCQ (Single Correct Answer)

Open solution

Which of the following reactions will yield benzaldehyde as a product?



(A) (B) and (C) **(B)** (C) and (D) **(C)** (A) and (D) **(D)** (A) and (C)

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Q9. MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement I : In Hofmann degradation reaction, the migration of only an alkyl group takes place from carbonyl carbon of the amide to the nitrogen atom.

Statement II : The group is migrated in Hofmann degradation reaction to electron deficient atom.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both Statement I and Statement II are correct.

(B) Both Statement I and Statement II are incorrect.

(C) Statement I is correct but Statement II is incorrect.

(D) Statement I is incorrect but Statement II is correct.

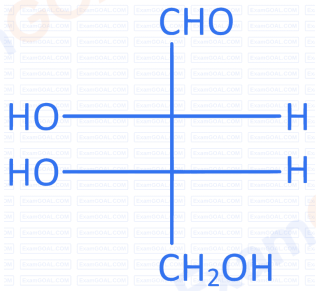
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Q10. MCQ (Single Correct Answer)

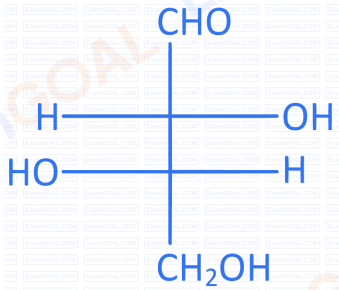
Open solution

L-isomer of a compound 'A' ($C_4H_8O_4$) gives a positive test with $[Ag(NH_3)_2]^+$. Treatment of 'A' with acetic anhydride yields triacetate derivative. Compound 'A' produces an optically active compound (B) and an optically inactive compound (C) on treatment with bromine water and HNO_3 respectively. Compound (A) is :

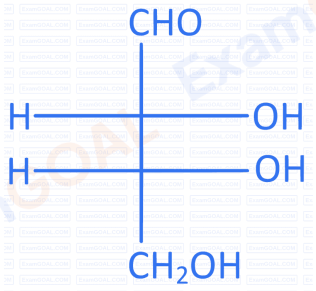
(A)



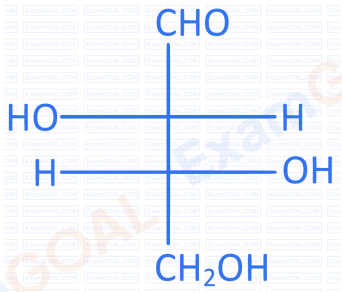
(B)



(C)



(D)



Q11. Numerical

Open solution

If the uncertainty in velocity and position of a minute particle in space are, 2.4×10^{-26} (m s^{-1}) and 10^{-7} (m) respectively. The mass of the particle in g is _____. (Nearest integer)

(Given : h = 6.626×10^{-34} Js)

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Q12. Numerical

2 g of a non-volatile non-electrolyte solute is dissolved in 200 g of two different solvents A and B whose ebullioscopic constants are in the ratio of 1 : 8. The elevation in boiling points of A and B are in the ratio $\frac{x}{y}$ (x : y). The value of y is _____. (Nearest integer)

Open solution

Q13. Numerical

$2NOCl(g) \rightleftharpoons 2NO(g) + Cl_2(g)$

In an experiment, 2.0 moles of NOCl was placed in a one-litre flask and the concentration of NO after equilibrium established, was found to be 0.4 mol/L. The equilibrium constant at 30°C is _____ $\times 10^{-4}$.

Open solution

Q14. Numerical

The limiting molar conductivities of NaI, $NaNO_3$ and $AgNO_3$ are 12.7, 12.0 and 13.3 mS m^2 mol^{-1}, respectively (all at 25°C). The limiting molar conductivity of AgI at this temperature is _____ mS m^2 mol^{-1}.

Open solution

Q15. Numerical

The rate constant for a first order reaction is given by the following equation:

$\ln k = 33.24 - \frac{2.0\times10^4\ K}{T}$

The activation energy for the reaction is given by _____ kJ mol^{-1}. (In nearest integer) (Given : R = 8.3 J K^{-1} mol^{-1})

Open solution

Q16. Numerical

The number of statements correct from the following for Copper (at. no. 29) is/are _____.

(A) Cu(II) complexes are always paramagnetic.

(B) Cu(I) complexes are generally colourless

(C) Cu(I) is easily oxidized

(D) In Fehling solution, the active reagent has Cu(I)

Open solution

Q17. Numerical

Acidified potassium permanganate solution oxidises oxalic acid. The spin-only magnetic moment of the manganese product formed from the above reaction is _____ B.M. (Nearest integer)

Open solution

Q18. Numerical

Two elements A and B which form 0.15 moles of A_2B and AB_3 type compounds. If both A_2B and AB_3 weigh equally, then the atomic weight of A is _____ times of atomic weight of B.

Open solution

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Q19. Numerical

Total number of possible stereoisomers of dimethyl cyclopentane is _____.

Open solution

Mathematics 22 questions

Q20. MCQ (Single Correct Answer)

The area of the polygon, whose vertices are the non-real roots of the equation $\overline{z} = iz^2$ is :

(A) $\frac{3\sqrt{3}}{4}$ (B) $\frac{3\sqrt{3}}{2}$ (C) $\frac{3}{2}$ (D) $\frac{3}{4}$

Open solution

Q21. MCQ (Single Correct Answer)

Let the system of linear equations
$x + 2y + z = 2$,
$\alpha x + 3y - z = \alpha$,
$-\alpha x + y + 2z = -\alpha$
be inconsistent. Then α is equal to :

(A) $\frac{5}{2}$ (B) $-\frac{5}{2}$ (C) $\frac{7}{2}$ (D) $-\frac{7}{2}$

Open solution

Q22. MCQ (Single Correct Answer)

$x = \sum_{n=0}^{\infty} a^n, y = \sum_{n=0}^{\infty} b^n, z = \sum_{n=0}^{\infty} c^n$, where a, b, c are in A.P. and |a| < 1, |b| < 1, |c| < 1, abc $\neq$ 0, then :

(A) x, y, z are in A.P.

(B) x, y, z are in G.P.

(C) $\frac{1}{x}, \frac{1}{y}, \frac{1}{z}$ are in A.P.

(D) $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 1 - (a + b + c)$

Open solution

Q23. MCQ (Single Correct Answer)

Let a be an integer such that $\lim_{x \to 7} \frac{18 - [1-x]}{[x-3a]}$ exists, where [t] is greatest integer $\leq$ t. Then a is equal to :

(A) -6 (B) -2 (C) 2 (D) 6

Open solution

Q24. MCQ (Single Correct Answer)

The number of distinct real roots of $x^4 - 4x + 1 = 0$ is :

(A) 4 (B) 2 (C) 1 (D) 0

Open solution

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Q25. MCQ (Single Correct Answer)

If $\cos^{-1}\left(\frac{y}{2}\right) = \log_e\left(\frac{x}{5}\right)^5$, $|y| < 2$, then :

(A) $x^2y'' + xy' - 25y = 0$ (B) $x^2y'' - xy' - 25y = 0$

(C) $x^2y'' - xy' + 25y = 0$ (D) $x^2y'' + xy' + 25y = 0$

Open solution

Q26. MCQ (Single Correct Answer)

If $\int \frac{(x^2+1)e^x}{(x+1)^2}dx = f(x)e^x + C$, where C is a constant, then $\frac{d^3f}{dx^3}$ at x = 1 is equal to :

(A) $-\frac{3}{4}$ (B) $\frac{3}{4}$ (C) $-\frac{3}{2}$ (D) $\frac{3}{2}$

Open solution

Q27. MCQ (Single Correct Answer)

The value of the integral

$\int_{-2}^{2} \frac{|x^3+x|}{(e^{x|x|}+1)}dx$ is equal to :

(A) $5e^2$ (B) $3e^{-2}$ (C) 4 (D) 6

Open solution

Q28. MCQ (Single Correct Answer)

If $\frac{dy}{dx} + \frac{2^{x-y}(2^y-1)}{2^x-1} = 0$, x, y > 0, y(1) = 1, then y(2) is equal to :

(A) $2 + \log_2 3$ (B) $2 + \log_3 2$ (C) $2 - \log_3 2$ (D) $2 - \log_2 3$

Open solution

Q29. MCQ (Single Correct Answer)

In an isosceles triangle ABC, the vertex A is (6, 1) and the equation of the base BC is 2x + y = 4. Let the point B lie on the line x + 3y = 7. If (α, β) is the centroid of ΔABC, then $15(\alpha + \beta)$ is equal to :

(A) 39 (B) 41 (C) 51 (D) 63

Open solution

Q30. MCQ (Single Correct Answer)

Let the eccentricity of an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, a > b$, be $\frac{1}{4}$. If this ellipse passes through the point $\left(-4\sqrt{\frac{2}{5}}, 3\right)$, then $a^2 + b^2$ is equal to :

(A) 29 (B) 31 (C) 32 (D) 34

Open solution

Q31. MCQ (Single Correct Answer)

If two straight lines whose direction cosines are given by the relations $l + m - n = 0$, $3l^2 + m^2 + cnl = 0$ are parallel, then the positive value of c is :

(A) 6 (B) 4 (C) 3 (D) 2

Open solution

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Q32. MCQ (Single Correct Answer)

Let $\vec{a} = \hat{i} + \hat{j} - \hat{k}$ and $\vec{c} = 2\hat{i} - 3\hat{j} + 2\hat{k}$. Then the number of vectors $\vec{b}$ such that $\vec{b} \times \vec{c} = \vec{a}$ and $|\vec{b}| \in \{1, 2, ..., 10\}$ is :

(A) 0 (B) 1 (C) 2 (D) 3

Open solution

Q33. MCQ (Single Correct Answer)

Five numbers x_1, x_2, x_3, x_4, x_5 are randomly selected from the numbers 1, 2, 3, ..., 18 and are arranged in the increasing order $(x_1 < x_2 < x_3 < x_4 < x_5)$. The probability that $x_2 = 7$ and $x_4 = 11$ is :

(A) $\frac{1}{136}$ (B) $\frac{1}{72}$ (C) $\frac{1}{68}$ (D) $\frac{1}{34}$

Open solution

Q34. MCQ (Single Correct Answer)

The value of $\cos\left(\frac{2\pi}{7}\right) + \cos\left(\frac{4\pi}{7}\right) + \cos\left(\frac{6\pi}{7}\right)$ is equal to :

(A) -1 (B) $-\frac{1}{2}$ (C) $-\frac{1}{3}$ (D) $-\frac{1}{4}$

Open solution

Q35. MCQ (Single Correct Answer)

$\sin^{-1}\left(\sin\frac{2\pi}{3}\right) + \cos^{-1}\left(\cos\frac{7\pi}{6}\right) + \tan^{-1}\left(\tan\frac{3\pi}{4}\right)$ is equal to :

(A) $\frac{11\pi}{12}$ (B) $\frac{17\pi}{12}$
(C) $\frac{31\pi}{12}$ (D) $-\frac{3\pi}{4}$

Open solution

Q36. Numerical

Let f : R $\to$ R be a function defined by $f(x) = \frac{2e^{2x}}{e^{2x}+e}$. Then $f\left(\frac{1}{100}\right) + f\left(\frac{2}{100}\right) + f\left(\frac{3}{100}\right) + \ ..... \ + f\left(\frac{99}{100}\right)$ is equal to _____.

Open solution

Q37. Numerical

If the sum of all the roots of the equation

$e^{2x} - 11e^{x} - 45e^{-x} + \frac{81}{2} = 0$ is $\log_e p$, then p is equal to _____.

Open solution

Q38. Numerical

The number of ways, 16 identical cubes, of which 11 are blue and rest are red, can be placed in a row so that between any two red cubes there should be at least 2 blue cubes, is _____.

Open solution

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Q39. Numerical

If the coefficient of x^{10} in the binomial expansion of $\left(\frac{\sqrt{x}}{5^{\frac{1}{4}}}+\frac{\sqrt{5}}{x^{\frac{1}{3}}}\right)^{60}$ is $5^k \cdot l$, where l, k $\in$ N and l is co-prime to 5, then k is equal to _____.

Open solution

Q40. Numerical

Let
$A_1=\{(x,y):|x|\leq y^2,|x|+2y\leq 8\}$ and
$A_2=\{(x,y):|x|+|y|\leq k\}$. If 27 (Area A_1) = 5 (Area A_2), then k is equal to :

Open solution

Q41. Numerical

A rectangle R with end points of one of its sides as (1, 2) and (3, 6) is inscribed in a circle. If the equation of a diameter of the circle is 2x $-$ y + 4 = 0, then the area of R is _____.

Open solution

Physics 26 questions

Q42. MCQ (Single Correct Answer)

A projectile is launched at an angle 'α' with the horizontal with a velocity 20 ms^{-1}. After 10 s, its inclination with horizontal is 'β'. The value of tanβ will be : (g = 10 ms^{-2}).

(A) tanα + 5secα
(B) tanα $-$ 5secα
(C) 2tanα $-$ 5secα
(D) 2tanα + 5secα

Open solution

Q43. MCQ (Single Correct Answer)

A girl standing on road holds her umbrella at 45° with the vertical to keep the rain away. If she starts running without umbrella with a speed of $15\sqrt{2}$ kmh^{-1}, the rain drops hit her head vertically. The speed of rain drops with respect to the moving girl is :

(A) 30 kmh^{-1}
(B) $\frac{25}{\sqrt{2}}$ kmh^{-1}
(C) $\frac{30}{\sqrt{2}}$ kmh^{-1}
(D) 25 kmh^{-1}

Open solution

Q44. MCQ (Single Correct Answer)

A silver wire has a mass (0.6 $\pm$ 0.006) g, radius (0.5 $\pm$ 0.005) mm and length (4 $\pm$ 0.04) cm. The maximum percentage error in the measurement of its density will be :

(A) 4%
(B) 3%
(C) 6%
(D) 7%

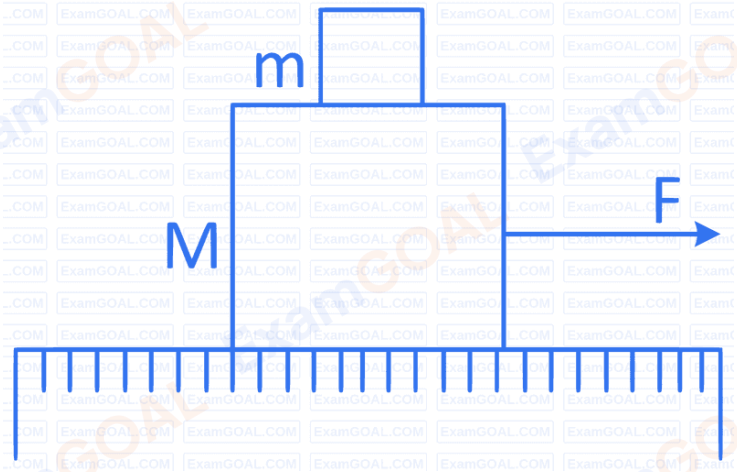
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Q45. MCQ (Single Correct Answer)

A system of two blocks of masses m = 2 kg and M = 8 kg is placed on a smooth table as shown in figure. The coefficient of static friction between two blocks is 0.5. The maximum horizontal force F that can be applied to the block of mass M so that the blocks move together will be :

Open solution



(A) 9.8 N (B) 39.2 N (C) 49 N (D) 78.4 N

Q46. MCQ (Single Correct Answer)

Two blocks of masses 10 kg and 30 kg are placed on the same straight line with coordinates (0, 0) cm and (x, 0) cm respectively. The block of 10 kg is moved on the same line through a distance of 6 cm towards the other block. The distance through which the block of 30 kg must be moved to keep the position of centre of mass of the system unchanged is :

Open solution

(A) 4 cm towards the 10 kg block (B) 2 cm away from the 10 kg block
(C) 2 cm towards the 10 kg block (D) 4 cm away from the 10 kg block

Q47. MCQ (Single Correct Answer)

A 72 Ω galvanometer is shunted by a resistance of 8 Ω. The percentage of the total current which passes through the galvanometer is :

Open solution

(A) 0.1% (B) 10% (C) 25% (D) 0.25%

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Q48. MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement I : The law of gravitation holds good for any pair of bodies in the universe.

Statement II : The weight of any person becomes zero when the person is at the centre of the earth.

In the light of the above statements, choose the correct answer from the options given below.

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true but Statement II is false

(D) Statement I is false but Statement II is true

Q49. MCQ (Single Correct Answer)

Open solution

What percentage of kinetic energy of a moving particle is transferred to a stationary particle when it strikes the stationary particle of 5 times its mass?

(Assume the collision to be head-on elastic collision)

(A) 50.0% (B) 66.6% (C) 55.6% (D) 33.3%

Q50. MCQ (Single Correct Answer)

Open solution

The velocity of a small ball of mass 'm' and density d_1, when dropped in a container filled with glycerin, becomes constant after some time. If the density of glycerin is d_2, then the viscous force acting on the ball, will be :

(A) $mg\left(1-\frac{d_1}{d_2}\right)$ (B) $mg\left(1-\frac{d_2}{d_1}\right)$

(C) $mg\left(\frac{d_1}{d_2}-1\right)$ (D) $mg\left(\frac{d_2}{d_1}-1\right)$

Q51. MCQ (Single Correct Answer)

Open solution

The susceptibility of a paramagnetic material is 99. The permeability of the material in Wb/A-m, is :

[Permeability of free space $\mu_0 = 4\pi \times 10^{-7}$ Wb/A-m]

(A) $4\pi \times 10^{-7}$ (B) $4\pi \times 10^{-4}$ (C) $4\pi \times 10^{-5}$ (D) $4\pi \times 10^{-6}$

Q52. MCQ (Single Correct Answer)

Open solution

The current flowing through an ac circuit is given by

I = 5 sin(120πt)A

How long will the current take to reach the peak value starting from zero?

(A) $\frac{1}{60}$ s (B) 60 s (C) $\frac{1}{120}$ s (D) $\frac{1}{240}$ s

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Q53. MCQ (Single Correct Answer)

Open solution

Match List-I with List-II :

	List - I		List - II
(a)	Ultraviolet rays	(i)	Study crystal structure
(b)	Microwaves	(ii)	Greenhouse effect
(c)	Infrared rays	(iii)	Sterilizing surgical instrument
(d)	X-rays	(iv)	Radar system

Choose the correct answer from the options given below :

(A) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)

(B) (a)-(iii), (b)-(i), (c)-(ii), (d)-(iv)

(C) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

(D) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

Q54. MCQ (Single Correct Answer)

Open solution

An α particle and a carbon 12 atom has same kinetic energy K. The ratio of their de-Broglie wavelengths $(\lambda_\alpha : \lambda_{C12})$ is :

(A) $1 : \sqrt{3}$ (B) $\sqrt{3} : 1$ (C) $3 : 1$ (D) $2 : \sqrt{3}$

Q55. MCQ (Single Correct Answer)

Open solution

A force of 10 N acts on a charged particle placed between two plates of a charged capacitor. If one plate of capacitor is removed, then the force acting on that particle will be.

(A) 5 N (B) 10 N (C) 20 N (D) Zero

Q56. MCQ (Single Correct Answer)

Open solution

The displacement of simple harmonic oscillator after 3 seconds starting from its mean position is equal to half of its amplitude. The time period of harmonic motion is :

(A) 6 s (B) 8 s (C) 12 s (D) 36 s

Q57. MCQ (Single Correct Answer)

Open solution

Consider a light ray travelling in air is incident into a medium of refractive index $\sqrt{2n}$. The incident angle is twice that of refracting angle. Then, the angle of incidence will be :

(A) $\sin^{-1}\left(\sqrt{n}\right)$ (B) $\cos^{-1}\left(\sqrt{\frac{n}{2}}\right)$

(C) $\sin^{-1}\left(\sqrt{2n}\right)$ (D) $2\cos^{-1}\left(\sqrt{\frac{n}{2}}\right)$

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Q58. MCQ (Single Correct Answer)

Open solution

A hydrogen atom in its ground state absorbs 10.2 eV of energy. The angular momentum of electron of the hydrogen atom will increase by the value of :

(Given, Planck's constant = 6.6×10^{-34} Js).

(A) 2.10×10^{-34} Js

(B) 1.05×10^{-34} Js

(C) 3.15×10^{-34} Js

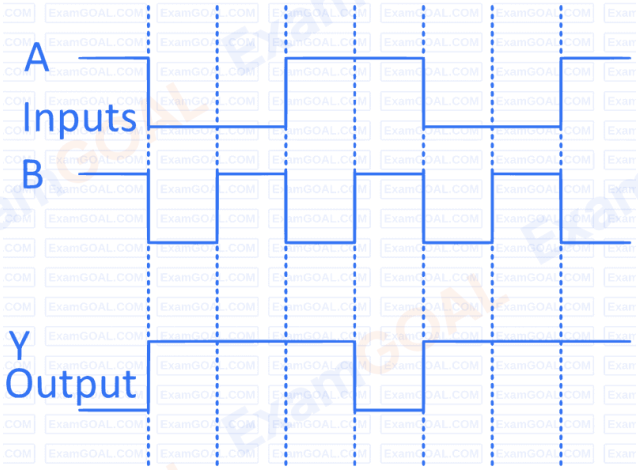
(D) 4.2×10^{-34} Js

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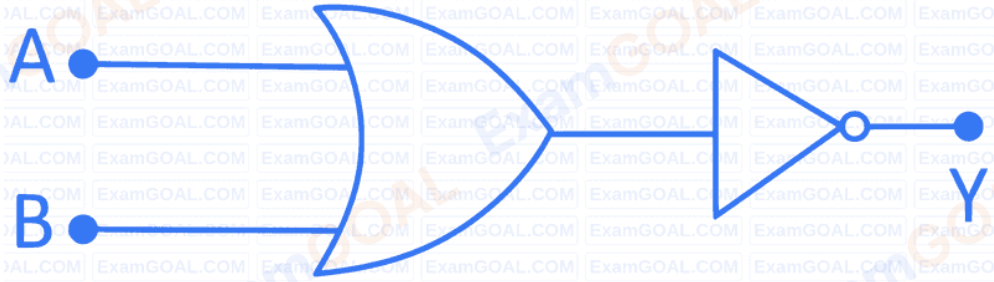
Q59. MCQ (Single Correct Answer)

Open solution

Identify the correct Logic Gate for the following output (Y) of two inputs A and B.



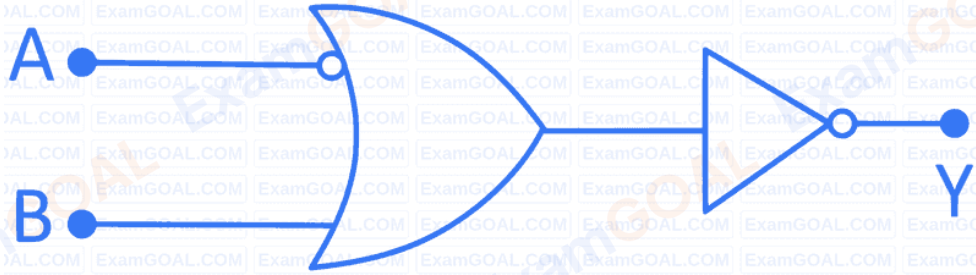
(A)



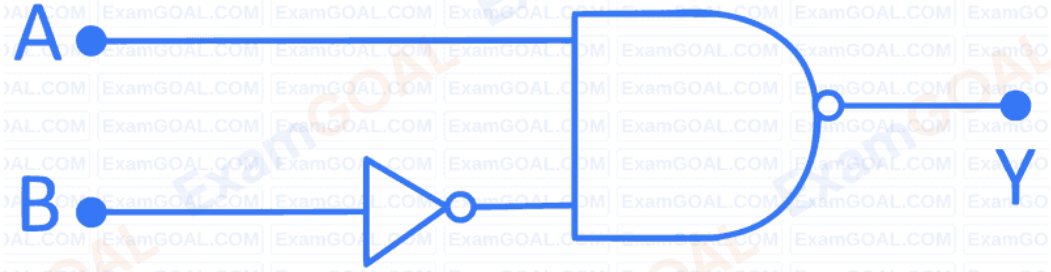
(B)



(C)



(D)



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Q60. MCQ (Single Correct Answer)

A mixture of hydrogen and oxygen has volume 2000 cm^3, temperature 300 K, pressure 100 kPa and mass 0.76 g. The ratio of number of moles of hydrogen to number of moles of oxygen in the mixture will be:

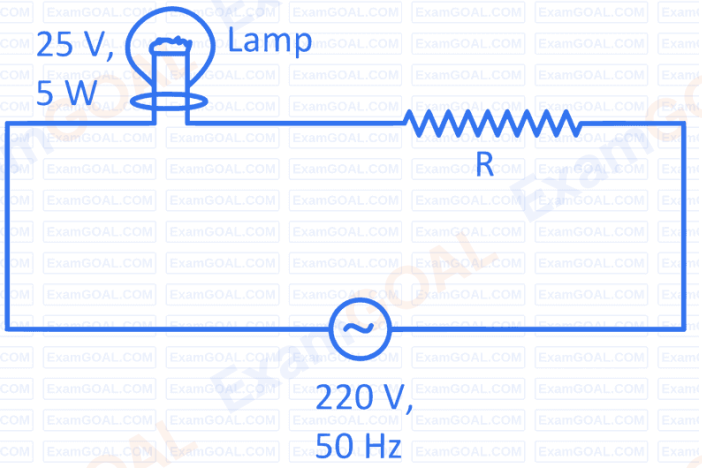
[Take gas constant R = 8.3 $JK^{-1}mol^{-1}$]

(A) $\frac{1}{3}$ (B) $\frac{3}{1}$ (C) $\frac{1}{16}$ (D) $\frac{16}{1}$

Open solution

Q61. Numerical

A 220 V, 50 Hz AC source is connected to a 25 V, 5 W lamp and an additional resistance R in series (as shown in figure) to run the lamp at its peak brightness, then the value of R (in ohm) will be _____.



Open solution

Q62. Numerical

In Young's double slit experiment the two slits are 0.6 mm distance apart. Interference pattern is observed on a screen at a distance 80 cm from the slits. The first dark fringe is observed on the screen directly opposite to one of the slits. The wavelength of light will be _____ nm.

Open solution

Q63. Numerical

A beam of monochromatic light is used to excite the electron in Li^{++} from the first orbit to the third orbit. The wavelength of monochromatic light is found to be $x \times 10^{-10}$ m. The value of x is _____.

[Given hc = 1242 eV nm]

Open solution

Q64. Numerical

The current density in a cylindrical wire of radius 4 mm is 4×10^6 Am^{-2}. The current through the outer portion of the wire between radial distances $\frac{R}{2}$ and R is _____ π A.

Open solution

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Q65. Numerical

A capacitor of capacitance 50 pF is charged by 100 V source. It is then connected to another uncharged identical capacitor. Electrostatic energy loss in the process is _____ nJ.

Open solution

Q66. Numerical

The area of cross-section of a large tank is 0.5 m^2. It has a narrow opening near the bottom having area of cross-section 1 cm^2. A load of 25 kg is applied on the water at the top in the tank. Neglecting the speed of water in the tank, the velocity of the water, coming out of the opening at the time when the height of water level in the tank is 40 cm above the bottom, will be _____ cms $^{-1}$. [Take g = 10 ms^{-2}]

Open solution

Q67. Numerical

A pendulum of length 2 m consists of a wooden bob of mass 50 g. A bullet of mass 75 g is fired towards the stationary bob with a speed v. The bullet emerges out of the bob with a speed $\frac{v}{3}$ and the bob just completes the vertical circle. The value of v is _____ ms^{-1}. (if g = 10 m/s^2).

Open solution

Out of syllabus 23 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q68. MCQ (Single Correct Answer) · **Chemistry**

Match List-I with List-II

	List - I		List - II
(A)	Lyophilic colloid	(I)	Liquid-liquid colloid
(B)	Emulsion	(II)	Protective colloid
(C)	Positively charged colloid	(III)	$FeCl_3$ + NaOH
(D)	Negatively charged colloid	(IV)	$FeCl_3$ + hot water

Open solution

Choose the correct answer from the options given below :

(A) (A) - (II), (B) - (I), (C) - (IV), (D) - (III)

(B) (A) - (III), (B) - (I), (C) - (IV), (D) - (II)

(C) (A) - (II), (B) - (I), (C) - (III), (D) - (IV)

(D) (A) - (III), (B) - (II), (C) - (I), (D) - (IV)

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Q69. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Match List-I with List-II

	List - I		List - II
(A)	Concentration of gold ore	(I)	Aniline
(B)	Leaching of alumina	(II)	NaOH
(C)	Froth stabiliser	(III)	SO_2
(D)	Blister copper	(IV)	NaCN

Choose the correct answer from the options given below :

(A) (A) - (IV), (B) - (III), (C) - (II), (D) - (I)

(B) (A) - (IV), (B) - (II), (C) - (I), (D) - (III)

(C) (A) - (III), (B) - (II), (C) - (I), (D) - (IV)

(D) (A) - (II), (B) - (IV), (C) - (III), (D) - (I)

Q70. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Addition of H_2SO_4 to BaO_2 produces :

(A) BaO, SO_2 and H_2O **(B)** $BaHSO_4$ and O_2 **(C)** $BaSO_4$, H_2 and O_2 **(D)** $BaSO_4$ and H_2O_2

Q71. MCQ (Single Correct Answer) · **Chemistry**

Open solution

$BeCl_2$ reacts with $LiAlH_4$ to give :

(A) Be + $Li[AlCl_4]$ + H_2

(B) Be + AlH_3 + LiCl + HCl

(C) BeH_2 + LiCl + $AlCl_3$

(D) BeH_2 + $Li[AlCl_4]$

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Q72. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Match List - I with List - II.

	List - I (Si-Compounds)		List - II (Si-Polymeric/Other Products
(A)	$(CH_3)_4Si$	(I)	Chain Silicone
(B)	$(CH_3)Si(OH)_3$	(II)	Dimeric Silicone
(C)	$(CH_3)_2Si(OH)_2$	(III)	Silane
(D)	$(CH_3)_3Si(OH)$	(IV)	2D - Silicone

Choose the correct answer from the options given below :

(A) (A) - (III), (B) - (II), (C) - (I), (D) - (IV)

(B) (A) - (IV), (B) - (I), (C) - (II), (D) - (III)

(C) (A) - (II), (B) - (I), (C) - (IV), (D) - (III)

(D) (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Q73. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Heating white phosphorus with conc. NaOH solution gives mainly :

(A) Na_3P and H_2O

(B) H_3PO and NaH

(C) $P(OH)_3$ and NaH_2PO_4

(D) PH_3 and NaH_2PO_2

Q74. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Given below are two Statements:

Statement I : Classical smog occurs in cool humid climate. It is a reducing mixture of smoke, fog and sulphur dioxide.

Statement II : Photochemical smog has components, ozone, nitric oxide, acrolein, formaldehyde, PAN etc.

In the light of the above statements, choose the most appropriate answer from the options given below.

(A) Both Statement I and Statement II are correct.

(B) Both Statement I and Statement II are incorrect.

(C) Statement I is correct but Statement II is incorrect.

(D) Statement I is incorrect but Statement II is correct.

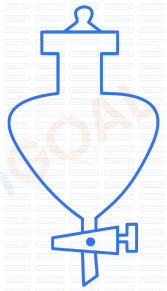
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Q75. MCQ (Single Correct Answer) · **Chemistry**

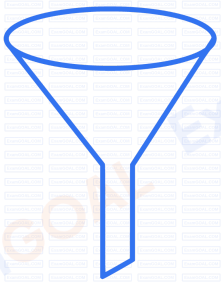
Open solution

Which of the following is structure of a separating funnel?

(A)



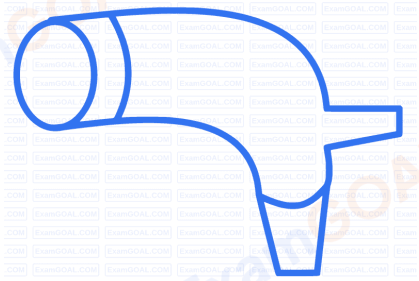
(B)



(C)



(D)



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Q76. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Match List - I with List - II

	List - I (Polymer)		**List - II (Used in)**
(A)	Bakelite	(I)	Radio and television cabinets
(B)	Glyptal	(II)	Electrical switches
(C)	PVC	(III)	Paints and Lacquers
(D)	Polystyrene	(IV)	Water pipes

Choose the correct answer from the options given below :

(A) (A) - (II), (B) - (III), (C) - (IV), (D) - (I)

(B) (A) - (I), (B) - (II), (C) - (III), (D) - (IV)

(C) (A) - (IV), (B) - (III), (C) - (II), (D) - (I)

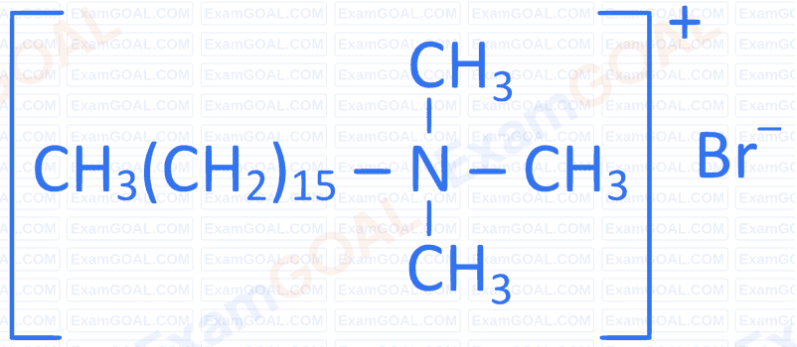
(D) (A) - (II), (B) - (III), (C) - (I), (D) - (IV)

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Q77. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Match List - I with List - II.

	List - I		List - II
(A)		(I)	Dishwashing powder
(B)		(II)	Toothpaste
(C)	$C_{17}H_{35}COO^{-}Na^{+} + Na_2CO_3$ + Rosinate	(III)	Laundry soap
(D)	$CH_3(CH_2)_{16}COO(CH_2CH_2O)_nCH_2CH_2OH$	(IV)	Hair conditioner

Choose the correct answer from the options given below :

(A) (A) - (III), (B) - (II), (C) - (IV), (D) - (I)

(B) (A) - (IV), (B) - (II), (C) - (III), (D) - (I)

(C) (A) - (IV), (B) - (III), (C) - (II), (D) - (I)

(D) (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Q78. Numerical · **Chemistry**

Open solution

Metal deficiency defect is shown by $Fe_{0.93}O$. In the crystal, some Fe^{2+} cations are missing and loss of positive charge is compensated by the presence of Fe^{3+} ions. The percentage of Fe^{2+} ions in the $Fe_{0.93}O$ crystals is _____. (Nearest integer)

Q79. MCQ (Single Correct Answer) · **Mathematics**

Open solution

Let $\frac{dy}{dx} = \frac{ax-by+a}{bx+cy+a}$, where a, b, c are constants, represent a circle passing through the point (2, 5). Then the shortest distance of the point (11, 6) from this circle is :

(A) 10 **(B)** 8 **(C)** 7 **(D)** 5

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Q80. MCQ (Single Correct Answer) · **Mathematics**

The lengths of the sides of a triangle are $10 + x^2$, $10 + x^2$ and $20 - 2x^2$. If for x = k, the area of the triangle is maximum, then $3k^2$ is equal to :

(A) 5 (B) 8 (C) 10 (D) 12

Open solution

Q81. MCQ (Single Correct Answer) · **Mathematics**

Let X be a random variable having binomial distribution B(7, p). If P(X = 3) = 5P(X = 4), then the sum of the mean and the variance of X is :

(A) $\frac{105}{16}$ (B) $\frac{7}{16}$ (C) $\frac{77}{36}$ (D) $\frac{49}{16}$

Open solution

Q82. MCQ (Single Correct Answer) · **Mathematics**

The boolean expression $(\sim (p \wedge q)) \vee q$ is equivalent to :

(A) $q \rightarrow (p \wedge q)$ (B) $p \rightarrow q$ (C) $p \rightarrow (p \rightarrow q)$ (D) $p \rightarrow (p \vee q)$

Open solution

Q83. Numerical · **Mathematics**

The positive value of the determinant of the matrix A, whose

Adj(Adj(A)) = $\begin{pmatrix} 14 & 28 & -14 \\ -14 & 14 & 28 \\ 28 & -14 & 14 \end{pmatrix}$, is _____.

Open solution

Q84. Numerical · **Mathematics**

If the sum of the first ten terms of the series

$\frac{1}{5} + \frac{2}{65} + \frac{3}{325} + \frac{4}{1025} + \frac{5}{2501} + \ldots$.

is $\frac{m}{n}$, where m and n are co-prime numbers, then m + n is equal to _____.

Open solution

Q85. Numerical · **Mathematics**

A circle of radius 2 unit passes through the vertex and the focus of the parabola $y^2 = 2x$ and touches the parabola $y = \left(x - \frac{1}{4}\right)^2 + \alpha$, where $\alpha > 0$. Then $(4\alpha - 8)^2$ is equal to _____.

Open solution

Q86. Numerical · **Mathematics**

Let the mirror image of the point (a, b, c) with respect to the plane $3x - 4y + 12z + 19 = 0$ be $(a - 6, \beta, \gamma)$. If $a + b + c = 5$, then $7\beta - 9\gamma$ is equal to _____.

Open solution

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Q87. MCQ (Single Correct Answer) · **Physics**

An observer moves towards a stationary source of sound with a velocity equal to one-fifth of the velocity of sound. The percentage change in the frequency will be :

(A) 20% (B) 10% (C) 5% (D) 0%

Open solution

Q88. Numerical · **Physics**

In a carnot engine, the temperature of reservoir is 527°C and that of sink is 200 K. If the work done by the engine when it transfers heat from reservoir to sink is 12000 kJ, the quantity of heat absorbed by the engine from reservoir is _____ $\times 10^6$ J.

Open solution

Q89. Numerical · **Physics**

A cell, shunted by a 8 Ω resistance, is balanced across a potentiometer wire of length 3 m. The balancing length is 2 m when the cell is shunted by 4 Ω resistance. The value of internal resistance of the cell will be _____ Ω.

Open solution

Q90. Numerical · **Physics**

The height of a transmitting antenna at the top of a tower is 25 m and that of receiving antenna is, 49 m. The maximum distance between them, for satisfactory communication in LOS (Line-Of-Sight) is $K\sqrt{5} \times 10^2$ m. The value of K is _____.

(Assume radius of Earth is $64 \times 10^{+5}$ m) [Calculate upto nearest integer value]

Open solution

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Answer Key

Chemistry

1. **A**	2. **B**	3. **B**	4. **D**	5. **C**
6. **D**	7. **A**	8. **C**	9. **D**	10. **A**
11. **22**	12. **8**	13. **125**	14. **14**	15. **166**
16. **3**	17. **6**	18. **2**	19. **6**	

Mathematics

20. **A**	21. **D**	22. **C**	23. **A**	24. **B**
25. **D**	26. **B**	27. **D**	28. **D**	29. **C**
30. **B**	31. **A**	32. **A**	33. **C**	34. **B**
35. **A**	36. **99**	37. **45**	38. **56**	39. **5**
40. **6**	41. **16**			

Physics

42. **B**	43. **C**	44. **A**	45. **C**	46. **C**
47. **B**	48. **A**	49. **C**	50. **B**	51. **C**
52. **D**	53. **A**	54. **B**	55. **A**	56. **D**
57. **D**	58. **B**	59. **B**	60. **B**	61. **975**
62. **450**	63. **114**	64. **48**	65. **125**	66. **300**
67. **10**				

Out of syllabus

68. **A**	69. **B**	70. **D**	71. **C**	72. **D**
73. **D**	74. **A**	75. **A**	76. **A**	77. **B**
78. **85**	79. **B**	80. **C**	81. **C**	82. **D**
83. **14**	84. **276**	85. **63**	86. **137**	87. **A**
88. **16**	89. **8**	90. **192**		

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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